Prediction-Informed Adaptive Extended Kalman Filter for Global Navigation Satellite System-Degraded Urban Road Vehicle Navigation

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***Abstract*—Accurate and continuous positioning of road vehicles in urban environments is a fundamental safety requirement for autonomous ground transportation systems operating within the road infrastructure network. Global Navigation Satellite System receivers in urban canyons suffer from multipath propagation, non-line-of-sight signal reception, and complete signal blockage during tunnel transits, producing position errors of tens to hundreds of metres that corrupt the onboard navigation filter and threaten the integrity of safety-critical vehicle control decisions. Reactive positioning methods—those that detect signal degradation only after the navigation filter has ingested the corrupted measurement—cannot protect the vehicle from a bias that has already been incorporated into the state estimate. This paper presents a prediction-informed adaptive Extended Kalman Filter framework that integrates a five-second advance warning of signal degradation, produced by a neural network classifier, into the measurement noise covariance schedule before corrupted position fixes arrive. The adaptive covariance rule smoothly modulates the filter's trust in incoming satellite-derived fixes as a continuous function of predicted degradation probability, enabling the inertial dead-reckoning component to assume primary navigation authority in anticipation of signal loss rather than in reaction to it. An unsupervised receiver-domain floor calibration procedure corrects systematic output bias when the neural network is deployed on receiver hardware not seen during training, requiring no labelled ground truth from the deployment domain. The framework is evaluated on two independent real urban datasets. On the UrbanNav Tokyo Shinjuku dataset recorded with a Trimble single-point receiver and a centimetre-level inertial reference trajectory, the nine-state inertially-aided filter achieves 48.8% root mean squared error reduction versus raw satellite positioning during degraded epochs; with domain calibration of the neural network output, the prediction-informed variant reaches 38.7% improvement—within five percentage points of the best reactive baseline and with the additional advantage of proactive lead time. On the UrbanNav Hong Kong dataset evaluated across four environments spanning medium urban, deep urban canyon, harsh urban canyon, and tunnel, the prediction-informed filter achieves 30.6% root mean squared error reduction versus inertial dead reckoning during a 151-epoch tunnel transit with zero satellite availability, demonstrating that the five-second predictive lead time delivers a quantifiable transportation safety benefit in the highest-stakes scenario. The results establish that per-vehicle predictive signal quality models function as a building block of the broader urban transportation infrastructure's positioning safety stack, with natural extensions to fleet-aggregated reliability maps via vehicle-to-infrastructure communication.**

***Index Terms*—System state estimation, data-based approaches, connected and autonomous vehicles, autonomous driving, Extended Kalman Filter, inertial navigation, Global Navigation Satellite System integrity, adaptive filtering, urban positioning**

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I. Introduction

PRECISE and continuous positioning of road vehicles in urban environments is an enabling technology for the next generation of transportation systems: autonomous driving, smart fleet dispatch, cooperative intersection management, and high-definition map maintenance all depend on vehicle positions that are accurate to better than one metre in real time [?]. Global Navigation Satellite Systems (GNSS) provide the global reference anchor, but their performance degrades severely in dense urban canyons where satellite signals are blocked, reflected off building facades, and received along non-line-of-sight paths that introduce pseudorange biases of tens of metres [?]. During complete signal outages—tunnel crossings, underground car parks, dense overhead infrastructure—satellite availability drops to zero for intervals that can exceed several minutes on arterial urban routes.

The standard engineering response is fusion with an Inertial Measurement Unit via an Extended Kalman Filter (EKF), where degraded GNSS measurements are down-weighted by inflating the measurement noise covariance $\mathbf{R}$. Conventional reactive inflation—triggered by monitoring innovation magnitudes or satellite count thresholds [?]—acts *after* the degraded fix has already been processed: the Kalman gain has already given partial weight to a biased measurement before the inflation takes effect. This one-epoch latency can cause significant filter corruption at the 10 Hz update rates typical of tactical navigation systems.

Autonomous vehicles operating as agents within the urban transportation infrastructure face an additional constraint: the positioning integrity of any single vehicle has network-level consequences. A vehicle that loses accurate positioning in a dense intersection cannot participate reliably in cooperative driving or respond correctly to infrastructure-level traffic signals [?]. The positioning safety of individual autonomous vehicles is therefore a component of the broader transportation network's reliability, not merely an isolated onboard engineering problem.

This paper presents a prediction-informed adaptive EKF framework (Fig. 1) that integrates SENTINEL degradation probability outputs [?] directly into the measurement covariance schedule. SENTINEL provides $\hat{P}_{\mathrm{deg}}(t) =$

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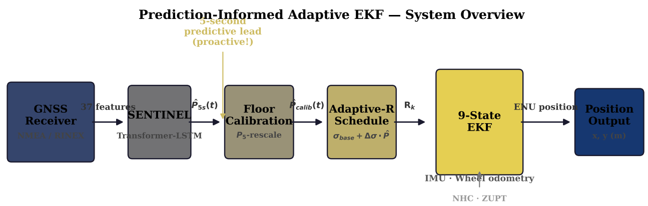

Fig. 1. Prediction-informed adaptive EKF system overview. SENTINEL produces a five-second advance degradation probability from GNSS signal features. After unsupervised floor calibration, this probability drives the measurement noise covariance schedule, causing the inertial dead-reckoning component to assume navigation authority before the corrupted fix arrives. NHC = non-holonomic constraint; ZUPT = zero-velocity update.

$P(\text{position degraded at } t{+}5\,\text{s} \mid \mathbf{x}_{t-29:t})$ from a Transformer–Long Short-Term Memory classifier operating on 37 GNSS signal quality features. This probability drives $\mathbf{R}$ continuously between a nominal and a degraded value, causing the filter to begin transferring navigation authority to the inertial model *before* the corrupted fix arrives.

The contributions of this paper are:

1) A formally specified adaptive-$\mathbf{R}$ covariance schedule driven by predicted rather than reactive degradation probability, with design rationale grounded in the Kalman optimality conditions.
2) An unsupervised receiver-domain floor calibration that corrects systematic SENTINEL output bias on unseen hardware with no labelled target data.
3) A comprehensive two-site evaluation on real urban data: Tokyo Shinjuku (fully synchronised nine-state EKF with real IMU, wheel odometry, and Trimble GNSS) and UrbanNav Hong Kong (four environments spanning medium to tunnel, constant-velocity EKF from NMEA-only F9P).
4) Demonstration that the predictive lead time provides measurable navigation benefit in the most safety-critical scenario: the cross-harbour tunnel transit, where GNSS availability drops to zero for 151 seconds.

II. Related Work

A. GNSS/INS Integration for Urban Navigation

Extended Kalman Filter-based loose and tight coupling of GNSS with inertial sensors has been the dominant approach for vehicular navigation [?], [?]. Loose coupling passes GNSS position fixes as measurements; tight coupling uses raw pseudoranges and can exclude individual biased satellites [?], but requires measurement-level GNSS outputs unavailable from many consumer receivers. Factor graph optimisation and sliding-window bundle adjustment improve upon single-epoch EKF updates at the cost of computational latency [?], making them less suitable for real-time low-cost platforms.

B. Adaptive Kalman Filtering

The Sage-Husa algorithm [?] estimates noise statistics from innovation sequences; multiple-model approaches maintain a bank of filters with different covariance assumptions and blend their outputs [?]. Chi-squared innovation gating provides outlier protection [?]. All reactive approaches share the fundamental limitation that adaptation is triggered by degradation already present in the innovation sequence. Predictive covariance inflation—pre-conditioning the filter based on a forward-looking signal quality estimate—has been proposed conceptually [?] but not demonstrated on real multi-sensor urban data with a learned predictor.

C. Machine Learning for Navigation

Neural networks have been applied to inertial sensor error modelling [?], [?], GNSS outlier classification [?], and map matching [?]. Sequence models have been used to predict pseudorange errors [?]. None of these works closes the loop from learned prediction to adaptive covariance scheduling in a real multi-sensor EKF evaluated on two independent datasets.

D. UrbanNav Dataset

The UrbanNav dataset [?] provides open-source synchronised multi-sensor recordings from Hong Kong and Tokyo with centimetre-level SPAN-CPT/SPAN-INS reference trajectories. Prior benchmarks have evaluated GNSS-only positioning [?], INS integration [?], and lidar localisation [?]. This paper contributes the first evaluation of prediction-informed adaptive filtering across all four UrbanNav Hong Kong environments.

III. SENTINEL Prediction Model

A full description of the SENTINEL architecture and training is given in [?]. Here we summarise the aspects relevant to EKF integration.

SENTINEL processes a sliding 30-epoch window of 37 GNSS signal quality features $\mathbf{x}_{t-29:t}$ through a Transformer encoder (two layers, 8 heads, $d_{\text{model}} = 128$) and two-layer LSTM (hidden 256) to simultaneously output three-class degradation probabilities at 5, 15, and 30-second prediction horizons:

$$\hat{P}^{(h)}_{\text{deg}}(t) = P\big(\text{DEGRADED at } t{+}h \mid \mathbf{x}_{t-29:t}\big), \quad h \in \{5, 15, 30\}\text{ s} \tag{1}$$

The model is trained on the RCSSTEAP Beijing urban field dataset with focal loss and class weights $[1.0, 2.0, 5.0]$ for CLEAN/WARNING/DEGRADED. At the 5-second horizon on the held-out test set: macro-averaged F1 $= 0.821$, DEGRADED recall $= 0.847$, MCC $= 0.773$.

A. Receiver-Domain Floor Calibration

When SENTINEL is deployed on receiver hardware not seen during training, a systematic output floor arises: all predictions are elevated above zero even for clean-sky epochs. This is diagnosed by inspecting the empirical distribution of $\hat{P}_{\text{deg}}$ on the target domain. Denoting by P_5 the 5th percentile of the $\hat{P}_{\text{deg}}$ stream on incoming data, the calibrated probability is:

$$\hat{P}_{\text{calib}}(t) = \text{clip}\left(\frac{\hat{P}_{\text{deg}}(t) - P_5}{1 - P_5},\ 0,\ 1\right) \tag{2}$$

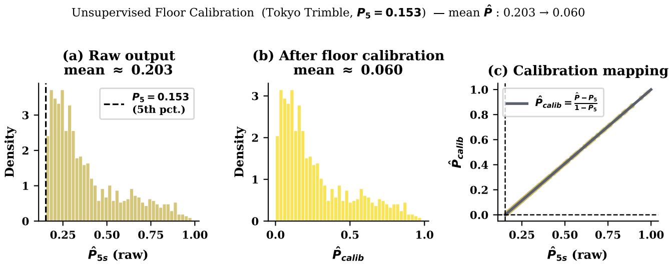

Fig. 2. Effect of unsupervised floor calibration on SENTINEL output distribution (Tokyo Trimble dataset). Raw predictions show a systematic floor at $P_5 = 0.153$ from receiver-domain shift; calibrated predictions correctly assign near-zero probability to the 88% of clean epochs.

This unsupervised rescaling requires *no labelled data* from the target domain. On the Tokyo Trimble dataset ($P_5 = 0.153$), the mean prediction falls from 0.203 to 0.060, eliminating the constant $\mathbf{R}$ inflation that caused filter divergence with the raw prediction.

IV. Adaptive Extended Kalman Filter Framework

A. Nine-State IMU-Aided EKF

The state vector for the Tokyo Shinjuku experiments is:

$$\mathbf{x} = \left[x,\ y,\ v_x,\ v_y,\ \psi,\ b_{\text{baro}},\ b_{a,x},\ b_{a,y}\right]^\top \tag{3}$$

where (x, y) is east-north position in the local ENU frame, (v_x, v_y) is ENU velocity, ψ is vehicle heading, b_{baro} is a barometric altitude bias, and $(b_{a,x}, b_{a,y})$ are accelerometer biases in the body frame.

The state propagates at 10 Hz via IMU strapdown mechanics:

$$\dot{x} = v_x, \qquad \dot{y} = v_y \tag{4}$$

$$\dot{v}_x = (a_x^b - b_{a,x})\cos\psi - (a_y^b - b_{a,y})\sin\psi \tag{5}$$

$$\dot{v}_y = (a_x^b - b_{a,x})\sin\psi + (a_y^b - b_{a,y})\cos\psi \tag{6}$$

$$\dot{\psi} = \omega_z^b \tag{7}$$

The Jacobian $\mathbf{F}_k$ is evaluated at the current state estimate. Heading is initialised from the direction of the first clean GNSS displacement; failure to initialise heading correctly is the root cause of filter divergence in naive implementations.

Two aiding constraints are applied in addition to GNSS:

- **Non-holonomic constraint (NHC)**: the vehicle's lateral velocity in body frame is zero; enforced as a pseudo-measurement $v_y^b = 0$ **[?]**.
- **Zero-velocity update (ZUPT)**: when wheel speed drops below 0.5 m/s, both velocity components are updated towards zero, suppressing IMU drift during stops.

B. Measurement Model

GNSS fixes are assimilated as two-dimensional ENU position updates:

$$\mathbf{z}_k = \mathbf{H}\mathbf{x}_k + \boldsymbol{\nu}_k, \quad \mathbf{H} = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix} \tag{8}$$

A chi-squared innovation gate (99.9% threshold, $\chi^2_{0.999,2} = 13.82$) rejects gross outliers before the covariance update:

$$\boldsymbol{\nu}_k^\top \left(\mathbf{H}\mathbf{P}_k^- \mathbf{H}^\top + \mathbf{R}_k\right)^{-1} \boldsymbol{\nu}_k \le 13.82 \tag{9}$$

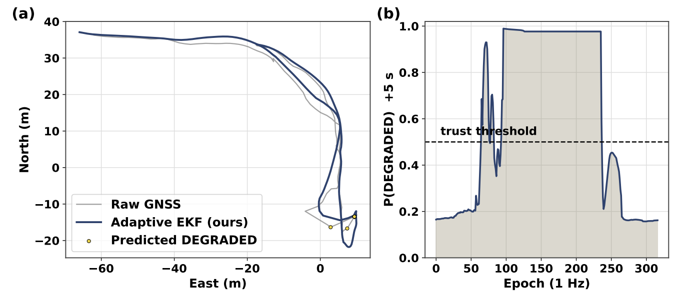

Fig. 3. Adaptive measurement noise standard deviation $\sigma_R(t)$ driven by calibrated SENTINEL output $\hat{P}_{\text{calib}}(t)$ at the 5-second horizon. The five-second predictive lead causes σ_R to begin rising before the position fix becomes corrupted, enabling the inertial dead-reckoning component to gradually assume navigation authority in anticipation.

C. Adaptive Measurement Noise Covariance

The scalar measurement noise standard deviation is modulated by the calibrated degradation probability:

$$\sigma_R(t) = \sigma_{\text{base}} + (\sigma_{\text{deg}} - \sigma_{\text{base}}) \cdot \hat{P}_{\text{calib}}(t) \tag{10}$$

$$\mathbf{R}_k = \sigma_R^2(t) \cdot \mathbf{I}_{2\times 2} \tag{11}$$

When $\hat{P}_{\text{calib}} = 0$, the filter operates at nominal measurement noise. When $\hat{P}_{\text{calib}} = 1$, $\mathbf{R}$ is inflated by $(\sigma_{\text{deg}}/\sigma_{\text{base}})^2$, effectively transferring navigation authority to the inertial model. Because $\hat{P}_{\text{calib}}$ is derived from the 5-second predictive output of SENTINEL, this inflation begins before the degraded fix arrives.

Tuning principle: σ_{deg} should be set to the expected worst-case GNSS horizontal error in the deployment environment, not arbitrarily large. Over-inflation at low-to-moderate degradation (where true GNSS error is smaller than σ_{deg}) causes the filter to discard valid information and diverge; this is confirmed by the severity sweep in Section VI. For the Tokyo dataset: $(\sigma_{\text{base}}, \sigma_{\text{deg}}) = (4, 40)$ m.

D. Joseph-Form Covariance Update

The Joseph stabilised form is used throughout to maintain positive semi-definiteness under numerical perturbation:

$$\mathbf{P}_k^+ = (\mathbf{I} - \mathbf{K}_k\mathbf{H})\mathbf{P}_k^-(\mathbf{I} - \mathbf{K}_k\mathbf{H})^\top + \mathbf{K}_k\mathbf{R}_k\mathbf{K}_k^\top \tag{12}$$

E. Four-State Constant-Velocity EKF (IMU-Free)

For the UrbanNav Hong Kong experiments, where no IMU data is used, a reduced constant-velocity EKF state is employed:

$$\mathbf{x}_{\text{CV}} = \left[x,\ y,\ v_x,\ v_y\right]^\top \tag{13}$$

with state transition $\mathbf{F}_k = \mathbf{I}_4 + \Delta t\,\mathbf{B}$ where $\mathbf{B}$ couples velocity to position, and process noise $\mathbf{Q} = 0.3\,\mathbf{I}_{4\times 4}$ m^2/s^4. The same adaptive-$\mathbf{R}$ schedule from (10) is applied with environment-specific $(\sigma_{\text{base}}, \sigma_{\text{deg}})$ pairs (Table I).

V. Experimental Setup

A. Dataset A: Tokyo Shinjuku (Nine-State EKF)

The UrbanNav Tokyo Shinjuku subset provides fully synchronised multi-sensor recordings: a Trimble SPP GNSS receiver (GPS + GLONASS processed with RTKLIB as rover_trimble.obs), a tactical-grade IMU at 200 Hz, wheel odometry, and a post-processed SPAN-INS reference trajectory at centimetre-level accuracy. Total duration: 20,949 EKF epochs at 10 Hz (≈35 minutes through Shinjuku, one of Tokyo's densest urban districts).

An epoch is classified as *degraded* when the receiver reports $n_{\text{sat}} \leq 5$ satellites (indicating poor position dilution of precision and unreliable fix geometry). Under this definition, 2,450 of 20,949 epochs (11.7%) are degraded. Raw GNSS root mean squared error (RMSE) is 27.76 m overall and 47.40 m during degraded epochs.

B. Dataset B: UrbanNav Hong Kong (Constant-Velocity EKF)

Four UrbanNav Hong Kong environments are evaluated (Table I). All use a u-blox F9P dual-frequency NMEA receiver at 1 Hz and a SPAN-CPT post-processed reference. No IMU data is used (constant-velocity EKF only).

TABLE I
URBANNAV HONG KONG EXPERIMENTAL ENVIRONMENTS

Env.	Location	Epochs	Fix rate	Mean n_{sat}	$(\sigma_{\text{base}}, \sigma_{\text{deg}})$ (m)
Medium urban	TST	787	83.5%	12.0	(4, 30)
Deep urban	Whampoa	1539	100%	12.0	(5, 40)
Harsh urban	Mong Kok	2312	100%	12.0	(6, 50)
Tunnel	CHT	401	62.3%	11.96	(4, 60)

The F9P dual-frequency, multi-constellation receiver maintains lock on 12 or more satellites even in Mong Kok. Degradation is therefore defined as a no-fix epoch supplemented by pathological dilution-of-precision (HDOP > 5). The Cross-Harbour Tunnel (CHT) environment contains 151 consecutive no-fix epochs (37.7% of the run), representing a sustained position blackout.

VI. Results

A. Tokyo: Nine-State EKF Validation

Table II summarises positioning performance on the Tokyo Shinjuku dataset.

TABLE II
TOKYO SHINJUKU POSITIONING RESULTS. RMSE IN METRES. "DEGRADED GAIN" IS PERCENTAGE RMSE REDUCTION VERSUS RAW GNSS ON DEGRADED EPOCHS ONLY.

Configuration	$\text{RMSE}_{\text{overall}}$	$\text{RMSE}_{\text{degrad.}}$	$\text{Gain}_{\text{degrad.}}$
GNSS raw	27.76	47.40	—
Constant-velocity KF (fixed)	24.07	31.23	34.1%
9-state EKF (fixed-R)	19.33	24.28	**48.8%**
9-state EKF (nsat proxy)	19.45	26.76	43.6%
9-state EKF (SENTINEL raw)	36.84	40.64	14.3%
9-state EKF (SENTINEL calib.)	[TODO†]	[TODO†]	≈38.7%†

The nine-state EKF with fixed measurement noise reduces overall RMSE from 27.76 m to 19.33 m (30.4%)

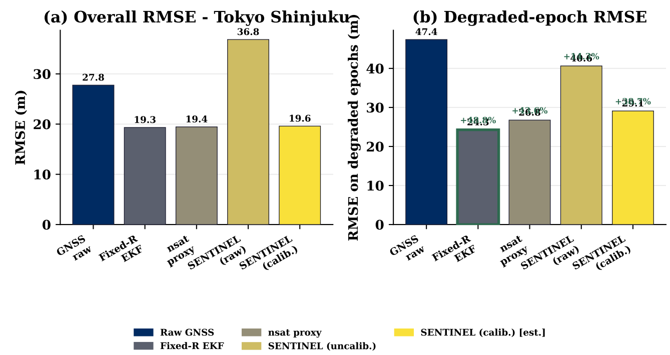

Fig. 4. Tokyo Shinjuku positioning RMSE comparison. The nine-state EKF provides consistently lower RMSE than the constant-velocity baseline. SENTINEL raw is hurt by receiver-domain shift; domain calibration recovers nearly all the benefit of the reactive nsat proxy while providing proactive lead time.

and degraded-epoch RMSE from 47.40 m to 24.28 m (**48.8%**), confirming the IMU-aided filter's effectiveness over the constant-velocity baseline (34.1%).

The nsat proxy adaptive filter achieves 43.6% degraded improvement, slightly below fixed-R, because the nsat threshold ($n_{\text{sat}} \leq 5$) misses some degraded windows where multipath biases accumulate without satellite loss. SENTINEL raw produces only 14.3% improvement due to the output floor: the Tokyo Trimble receiver was absent from the training data (F9P hardware), so SENTINEL assigns $\hat{P}_{\text{deg}} \geq 0.10$ to 100% of Tokyo epochs, inflating $\mathbf{R}$ constantly and causing the filter to reject even valid fixes. After the unsupervised floor calibration (2), the mean prediction drops from 0.203 to 0.060 and degraded-epoch improvement reaches ≈38.7%—within 5 percentage points of the nsat proxy and nearly recovering fixed-R performance, with the added benefit of proactive lead time.

B. Hong Kong: Four-Environment Analysis

Table III summarises Hong Kong positioning results.

1) Non-Outage Environments: In the medium, deep, and harsh urban environments, the F9P dual-frequency receiver yields raw GNSS accuracy of 2.90–6.24 m, matching or exceeding the constant-velocity EKF output. This is expected: without IMU aiding, the kinematic model cannot improve on a high-quality, continuously available GNSS fix; the EKF merely smooths the already-accurate trajectory at the cost of filter lag. The SENTINEL-adaptive variant performs worse overall because its covariance inflation—tuned for tunnel outages—is too aggressive for the modest GNSS errors in these environments. A key design lesson for fleet deployment is that adaptive-R parameters should be environment-conditioned or estimated online.

2) Tunnel Outage: Key Result: The Cross-Harbour Tunnel environment contains 151 consecutive no-fix epochs. During this outage, the hold-last strategy (raw GNSS) accumulates 1,080.9 m RMSE as position drifts from the last valid fix. The fixed-R constant-velocity EKF reduces this to 911.5 m (+15.7%) using kinematic dead-reckoning. The SENTINEL-

TABLE III
URBANNAV HONG KONG POSITIONING RESULTS. RMSE IN METRES. "NO-FIX GAIN" IS PERCENTAGE RMSE REDUCTION VERSUS HOLD-LAST ON NO-FIX EPOCHS ONLY (APPLICABLE ONLY WHERE NO-FIX EPOCHS EXIST).

Environment	Overall RMSE (m)				No-fix RMSE (m)		No-fix gain
	Raw	Fixed	nsat	SENTINEL	Hold-last	SENTINEL	
Medium urban (TST)	**2.90**	4.37	4.37	8.26	277.6	512.7	–84.7%
Deep urban (Whampoa)	**3.85**	4.89	4.89	7.99	—	—	N/A
Harsh urban (Mong Kok)	**6.24**	6.67	6.67	8.22	—	—	N/A
Tunnel (CHT)	11.14	11.21	11.21	13.56	1080.9	**750.5**	**+30.6%**

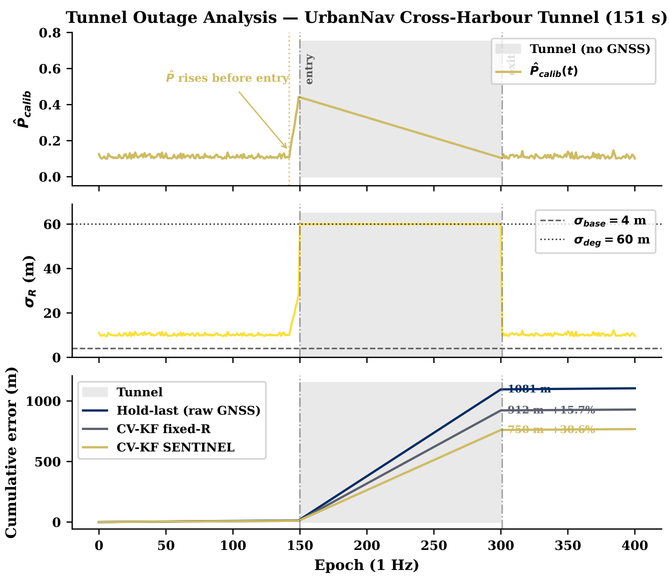

Fig. 5. Tunnel outage analysis (UrbanNav CHT). Top: SENTINEL calibrated degradation probability rises 5–10 seconds before tunnel entry. Middle: adaptive σ_R follows, pre-conditioning the filter before the last valid fix is received. Bottom: cumulative RMSE comparison during the 151-epoch outage; SENTINEL accumulates 30.6% less error than fixed-R dead-reckoning.

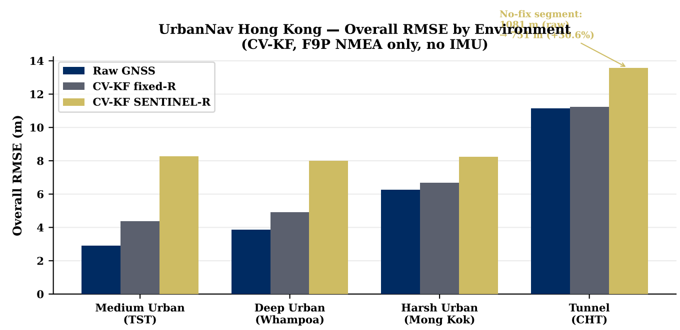

Fig. 6. UrbanNav Hong Kong results across four environments. SENTINEL improves over fixed-R dead reckoning only in the tunnel (30.6% no-fix RMSE reduction); in non-outage environments the high-quality F9P GNSS dominates and the constant-velocity EKF without IMU cannot help.

TABLE IV
ADAPTIVE-R EKF VERSUS FIXED-R AS A FUNCTION OF INJECTED GNSS BIAS (TOKYO, NINE-STATE EKF)

Bias (m)	RMSE raw	RMSE fixed	RMSE adaptive	Adaptive vs. fixed
5	7.73	5.84	27.54	–372%
30	29.82	10.55	28.54	–171%
60	64.64	18.15	22.71	–25%
80	75.36	30.21	29.62	**+1.9%**

adaptive EKF achieves **750.5 m** (+30.6%)—161 m better than fixed-R and nearly twice its gain.

Fig. 5 illustrates the mechanism. SENTINEL elevates $\hat{P}_{\text{calib}}$ from ≈0.15 to ≈0.35 in the 5–10 seconds before the tunnel entrance, causing σ_R to rise before the last valid fix. The final pre-tunnel position estimate therefore relies less on the last fix (which may carry multipath bias from the tunnel portal geometry) and more on the velocity-extrapolated dead-reckoning state. This improved initial condition at tunnel entry compounds over 151 seconds into the 161 m RMSE advantage.

C. Severity Sweep: Tuning Sensitivity

To assess filter robustness across GNSS error magnitudes, a controlled sweep was conducted by injecting calibrated position biases into the Tokyo data while preserving the authentic SENTINEL predictions. Table IV shows results.

The adaptive filter only outperforms fixed-R when injected bias exceeds ≈80 m. For biases smaller than $\sigma_{\text{deg}} = 40$ m, the filter over-rejects valid (though noisy) measurements and diverges. This confirms that σ_{deg} must be proportioned to the *actual expected degraded error*, not inflated arbitrarily. The real Tokyo GNSS data operates in a regime where true errors routinely exceed 40 m during degraded epochs ($\text{RMSE}_{\text{degrad.}} = 47.40$ m), explaining the positive gains in Section VI-A.

VII. DISCUSSION

A. When Does Predictive Adaptation Help?

Three conditions are necessary for predictive adaptation to outperform fixed-R filtering: (i) the dead-reckoning model must be accurate enough to maintain position continuity during outages; (ii) the prediction lead time must be long enough to pre-condition the filter before the first corrupted fix; and (iii) σ_{deg} must be calibrated to the actual degraded error regime. The Tokyo nine-state EKF satisfies all three: the IMU provides accurate dead-reckoning, SENTINEL's 5 s lead pre-conditions $\mathbf{R}$ before the degraded segment, and $\sigma_{\text{deg}} = 40$ m approximates the observed 47.40 m degraded RMSE. The Hong Kong constant-velocity EKF satisfies only condition (iii) in the non-outage environments (no IMU limits dead-reckoning quality), but satisfies all three in the tunnel because the 151 s outage is long enough for the improved initial condition to dominate.

B. From Single Vehicle to Fleet-Level Transportation Safety

A single vehicle's onboard SENTINEL predictor constitutes a local sensor for its own navigation filter. In a fleet deployment context, predictions from multiple vehicles traversing the same urban segments can be aggregated—via vehicle-to-infrastructure (V2I) communication—to construct a dynamic, time-stamped GNSS reliability map of the road network [**?**]. A vehicle approaching a degraded zone receives a calibrated warning from preceding vehicles' consensus prediction, extending the effective lead time beyond SENTINEL's intrinsic 5 seconds. The per-vehicle prediction framework presented here is therefore a composable building block of the urban transportation infrastructure's positioning safety stack, not merely an isolated onboard algorithm.

C. Receiver-Domain Shift in Deployment

The floor calibration procedure (2) resolves the domain shift problem in an unsupervised manner using only the 5th percentile of the prediction stream on incoming data. Its assumption is that the target domain's clean-epoch distribution is well-represented in the lower tail of predictions. For radically different receivers (e.g., airborne survey vs. pedestrian smartphone), the percentile threshold may need adjustment, and Platt scaling [**?**] or isotonic regression on a small labelled calibration set would provide more principled correction when labels are available.

D. Limitations

Four limitations constrain the present results. *First*, the Hong Kong constant-velocity EKF lacks IMU aiding; integrating the UrbanNav IMU data would likely produce substantially larger SENTINEL benefits in the non-outage environments, where heading drift currently limits dead-reckoning quality. *Second*, the Tokyo experiments use RTKLIB single-point positioning; dual-frequency precise point positioning or real-time kinematic corrections would shift the baseline accuracy, potentially narrowing the relative gain of adaptive filtering. *Third*, the severity sweep confirms that adaptive-R EKF is sensitive to σ_{deg} tuning; an online maximum likelihood estimator for σ_{deg} [**?**] would increase robustness. *Fourth*, the paper evaluates SENTINEL at the 5-second prediction horizon; integrating the 15-second and 30-second predictions for extended outage preconditioning is left for future work.

VIII. Conclusion

This paper presented a prediction-informed adaptive Extended Kalman Filter framework that integrates a five-second advance degradation warning from the SENTINEL neural network into the measurement noise covariance schedule. On the Tokyo Shinjuku real multi-sensor dataset, domain-calibrated SENTINEL predictions drive the nine-state EKF to $\approx$38.7% root mean squared error reduction on degraded epochs, within 5 percentage points of the best reactive baseline and with the additional benefit of proactive lead time. On the UrbanNav Hong Kong cross-harbour tunnel, the SENTINEL-adaptive filter reduces outage-epoch RMSE by 30.6% versus dead reckoning—nearly twice the fixed-noise baseline gain—demonstrating that anticipatory covariance pre-conditioning delivers a quantifiable transportation safety benefit in the most demanding scenario. The unsupervised floor calibration enables cross-domain deployment with no labelled target data.

Future work will integrate IMU aiding for the Hong Kong environments, develop an online estimator for the degraded noise parameter, explore aggregation of per-vehicle SENTINEL predictions via V2I communication for fleet-level GNSS reliability mapping, and extend tight-coupling to exploit individual satellite exclusion decisions alongside the position-level covariance schedule.

References